SRI HARSHINI

Data Engineer | AWS | Databricks | PySpark | Spark Streaming | Kafka | SQL | Lakehouse

Chicago, IL | sriharshiniraok@gmail.com | +1 (331) 701-1913

PROFESSIONAL SUMMARY

Data Engineer with 3.5+ years of experience developing scalable data pipelines, SQL-driven analytics datasets, data validation frameworks, and cloud-based lakehouse workflows across structured and semi-structured data. Hands-on experience with Python, SQL, PySpark, Apache Spark, Databricks, AWS S3, Delta Lake, relational databases, dimensional modeling, Git/GitHub, and production support of ETL/ELT data workflows. Skilled in building batch and streaming-oriented pipeline patterns, source-to-target reconciliation, schema evolution handling, data governance controls, and curated datasets for BI, analytics, and data science consumption. Strong Computer Science background with practical experience collaborating with application teams, QA, business users, and IT stakeholders to troubleshoot data issues and deliver reliable information products.

TECHNICAL SKILLS

Cloud & Lakehouse	AWS S3, AWS Glue, CloudWatch, IAM/RBAC, Azure Databricks, Azure Data Factory, ADLS Gen2, Delta Lake, Bronze/Silver/Gold Lakehouse, Parquet, JSON, CSV, Data Lakes, Data Warehouses, Data Marts
Big Data & Streaming	Apache Spark, PySpark, Spark SQL, Spark Structured Streaming, Spark Streaming concepts, Kafka concepts, Databricks notebooks/jobs, Hadoop, Hive, batch processing, incremental loads, pipeline monitoring
Programming & SQL	Python, SQL, T-SQL, PySpark, Java, Shell Scripting, Bash, PowerShell, REST APIs, JSON/XML parsing, reusable data utilities, automation scripts
Databases & Modeling	PostgreSQL, MySQL, SQL Server, Azure SQL Database, MongoDB/NoSQL concepts, relational schemas, schema evolution, Star Schema, Snowflake Schema, Fact/Dimension tables, SCD Type 1/2
Data Engineering	ETL/ELT, data ingestion, data transformation, data cleansing, source-to-target mapping, data quality rules, anomaly checks, reconciliation, audit logging, metadata documentation, data versioning concepts
Tools & SDLC	Git, GitHub, Jira, ServiceNow, Agile/Scrum, CI/CD concepts, Docker basics, unit testing concepts, VS Code, SSMS, Postman, Power BI, Tableau concepts, Excel

PROFESSIONAL EXPERIENCE

Junior Data Engineer | Tuve Group | Telangana, India **Jun 2021 - Dec 2023**

- Supported design, development, and maintenance of scalable data ingestion and ETL/ELT workflows for business reporting, customer analytics, and operational data platforms using SQL, Python, PySpark, and relational databases.
- Built SQL transformations, joins, views, stored procedures, staging tables, reporting tables, and reusable scripts to prepare curated datasets for downstream analytics and BI consumption.
- Assisted senior engineers in developing PySpark transformation logic to clean, join, aggregate, and standardize structured and semi-structured data from databases, CSV, Excel, JSON files, and application exports.
- Supported Databricks-style lakehouse patterns by organizing raw, cleansed, and analytics-ready datasets into bronze, silver, and gold layers with partitioned files and documented transformation rules.
- Created data quality checks for duplicate records, null handling, invalid formats, schema mismatches, referential accuracy, record counts, and source-to-target reconciliation before releasing data to reporting users.
- Worked on incremental load logic and schema evolution handling by reviewing source field changes, target table definitions, lookup rules, and exception records to reduce pipeline failures and reporting defects.
- Developed Python utilities for file validation, data profiling, formatting checks, repetitive data preparation, exception report generation, and basic automation of recurring data engineering tasks.
- Supported Kafka/Spark Streaming learning and prototype activities for ingesting event-style JSON data, validating message structures, and preparing near-real-time data extracts for analytics use cases.
- Collaborated with application teams, QA teams, business users, and senior engineers to investigate ETL failures, SQL errors, performance bottlenecks, data mismatches, file load issues, and root-cause patterns.
- Optimized SQL queries by reviewing joins, filters, indexes, aggregate logic, temporary tables, and execution patterns to improve recurring reporting and analytical extraction performance.
- Followed Git/GitHub version control practices, Agile sprint tasks, Jira defect tracking, peer review feedback, and deployment documentation to support a disciplined SDLC for data pipeline changes.
- Prepared technical documentation for source-to-target mappings, ETL workflows, data validation rules, data handling procedures, issue resolution steps, access control notes, and recurring production support activities.

Environment: AWS S3, SQL Server, MySQL, PostgreSQL, Python, PySpark, Apache Spark, Spark SQL, Databricks concepts, ETL/ELT, data lakehouse, CSV, JSON, Parquet, Git, GitHub, Jira, Agile, Windows, Linux basics

Technical Support Intern / Student Technical Assistant | Lewis University | Romeoville, IL **Jul 2024 - Jun 2025**

- Supported university technology operations while pursuing M.S. Computer Science, combining user support, SQL reporting, data validation, documentation, and internal system troubleshooting.
- Developed SQL queries to retrieve, validate, and analyze operational records from university systems for administrative reporting, audits, enrollment tracking, student services, and departmental requests.
- Prepared structured datasets from Excel files, CSV files, database tables, and system exports; identified missing values, duplicates, invalid formats, and inconsistent student, course, faculty, and departmental records.
- Automated recurring data preparation and reporting support tasks using SQL, Python, and Excel-based workflows, reducing repetitive manual review and improving report consistency.
- Supported user account and access-management activities including password resets, login troubleshooting, permission verification, and documentation of access-related procedures.
- Worked with university IT staff to troubleshoot application issues, validate system records, resolve data discrepancies, document standard operating procedures, and communicate findings clearly to non-technical users.

Environment: SQL, Python, Excel, CSV, Database Systems, Reporting Data, Windows, Helpdesk Tools, University Systems, Data Validation, Data Cleaning, Documentation, User Access Support

PROJECT EXPERIENCE

AWS Databricks Lakehouse Pipeline for Marketing and Customer Analytics

- Designed an end-to-end lakehouse workflow using AWS S3, Databricks, PySpark, Delta Lake concepts, and SQL to ingest, cleanse, transform, and publish structured and semi-structured datasets for analytics teams.
- Implemented bronze, silver, and gold layer design to separate raw ingestion, validated business data, and curated information products used by BI and data science consumers.
- Developed PySpark transformations for JSON/CSV parsing, data type standardization, deduplication, joins, aggregations, surrogate-key preparation, and fact/dimension table loading.
- Added data quality and anomaly checks for record counts, null thresholds, duplicate keys, schema drift, invalid date formats, referential checks, and source-to-target reconciliation with exception outputs.
- Documented governance rules for dataset ownership, access control, audit columns, metadata tracking, versioned processing logic, and reproducible pipeline reruns.

Spark Streaming and Kafka Event Processing Prototype

- Built a prototype event-processing pipeline using Kafka concepts, Spark Structured Streaming/PySpark, JSON event payloads, checkpointing concepts, and micro-batch processing patterns.
- Validated incoming event records, handled malformed payloads, applied basic enrichment logic, and wrote analytics-ready outputs for downstream dashboards and pipeline monitoring use cases.
- Created reusable Python validation functions to accelerate root-cause analysis, automate routine data quality checks, and identify recurring ingestion issues during development testing.

Sales and Customer Analytics Data Mart

- Built dimensional models using customer, product, location, and date dimensions with sales and activity fact tables to support KPI reporting and business analysis.
- Developed SQL queries, views, aggregations, and reconciliation checks to confirm accuracy between source extracts and reporting tables while improving query readability and maintainability.

EDUCATION

Lewis University, Romeoville, IL	Master of Science in Computer Science | Jan 2024 - Dec 2025 Relevant Coursework: Database Systems, Software Engineering, Software Design and Architecture
Vignan's Institute of Management and Technology for Women, Telangana, India	Bachelor of Technology in Information Technology | Aug 2019 - Jul 2023 Relevant Coursework: Java, Operating Systems, Database Management Systems, Software Engineering

PROFESSIONAL DEVELOPMENT

Databricks Lakehouse Fundamentals - In Progress | Microsoft Azure Data Engineer Associate - Planned | AWS Cloud Practitioner - Planned